

Regression Analysis Tutorial

Prescribed Text Books & Links

Montgomery, D. C., G.C. **Runger**, *Applied Statistics and Probability for Engineers*. 5th ed. New Delhi: Wiley-India, 2011.

Draper, N. R., H. Smith, *Applied Regression Analysis*. 3rd ed. New Delhi: Wiley-India, 1998.

Kutner, M. H., C. J. Nachtschiem, J. Netner, *Applied Linear Regression Models*. 4th ed. New Delhi: McGraw Hill, 2004.

Note: *In case of any reference be inadvertently omitted, the error will be rectified if intimated*

Interesting web links: <https://www.kaggle.com/code/ashydv/housing-price-prediction-linear-regression>

Warming Up Problem: A Short Story

- ❖ In a renovated airport, the management wants to address the complaint that waiting time is very long for the luggage conveyor belt
- ❖ So, the obvious solution is to increase the belt speed
- ❖ But, the management is afraid that increasing the speed may reduce the number of luggages removed in one run
- ❖ So an experiment is conducted to find optimum conveyor speed

Results from the Experiment

Sl. No.	Conveyor Belt Speed (cm/s)	Number of luggages picked up/minute
1	10	4
2	10	5
3	11	7
4	12	11
5	13	12
6	14	14
7	15	17
8	16	18
9	17	16
10	18	15
11	20	12

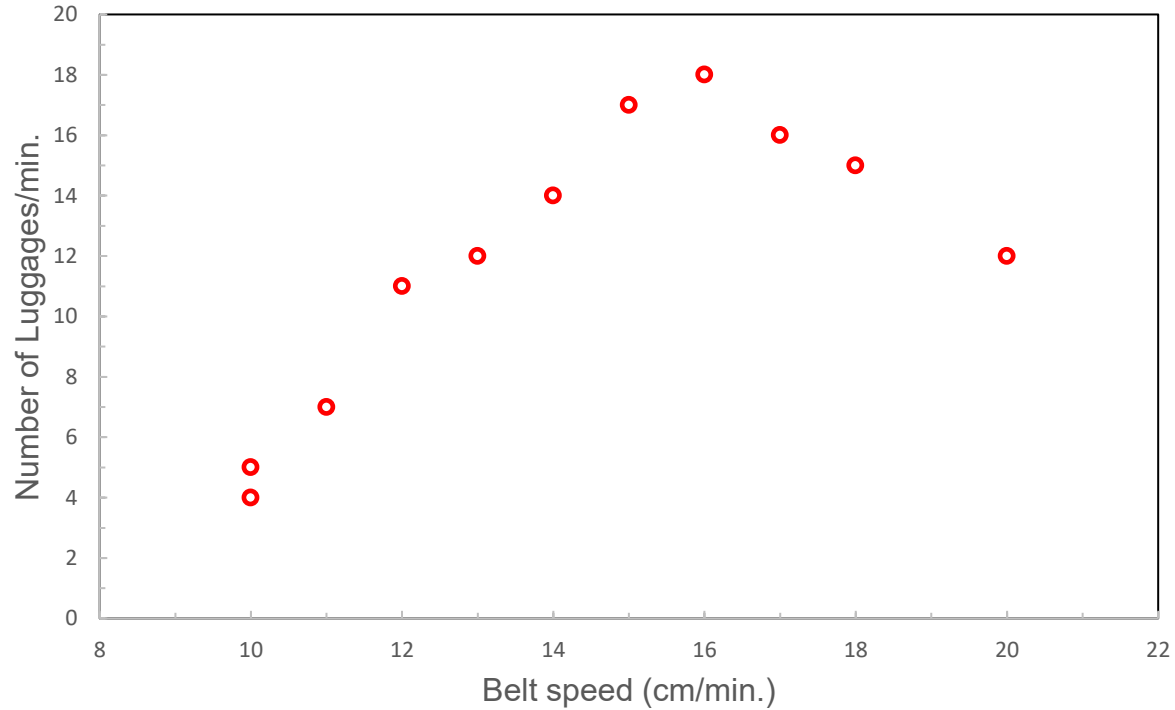
Data Scrutiny

- ❖ Number of Predictors:
- ❖ Number of Responses:
- ❖ How does the number of luggages pickedup per minute depend on conveyor belt speed?
- ❖ Obviously optimum is 16 cm/s so why do regression?

Data Scrutiny

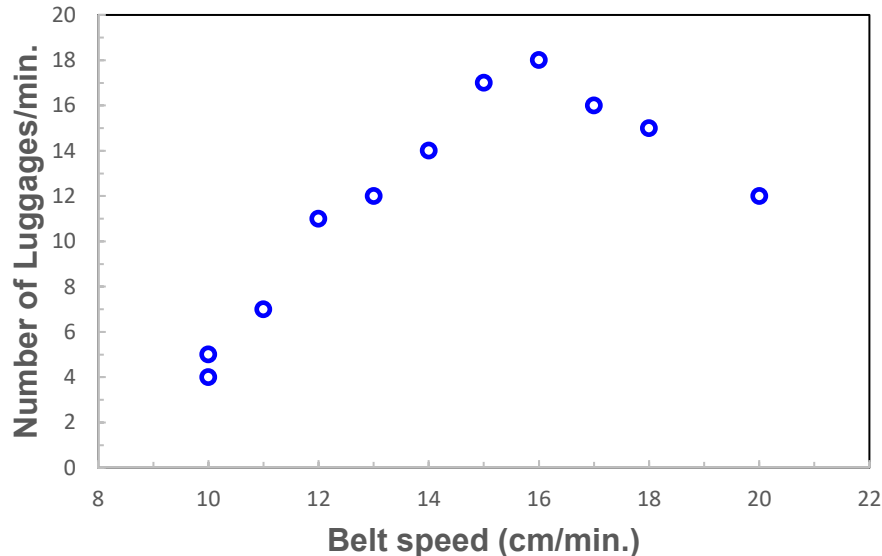
- ❖ Number of Predictors: 1 (conveyor belt speed)
- ❖ Number of Responses: 1
- ❖ How does the number of luggages picked up per minute depend on conveyor belt speed? Increases and decreases beyond a critical conveyor belt speed
- ❖ Obviously optimum is 16 cm/s so why do regression?
 - a) We do not know exact optimum
 - b) We may need to find conveyor belt speed at say 14.75 cm/s as it may be a safety limit constraint

Data Plotting



Model Choice

- ❖ What linear regression models may be tried?
 - a) First order model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X_1$ (obviously looks inadequate)
 - b) Second order model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2^2$ (may be tried)



Setting up the X Matrix (II Order Model)

Sl. No.	Conveyor Belt Speed (cm/s)	Number of luggages picked up/minute
1	10	4
2	10	5
3	11	7
4	12	11
5	13	12
6	14	14
7	15	17
8	16	18
9	17	16
10	18	15
11	20	12

$$\mathbf{X} = \begin{bmatrix} 1 & 10 & 100 \\ 1 & 10 & 100 \\ 1 & 11 & 121 \\ 1 & 12 & 144 \\ 1 & 13 & 169 \\ 1 & 14 & 196 \\ 1 & 15 & 225 \\ 1 & 16 & 256 \\ 1 & 17 & 289 \\ 1 & 18 & 324 \\ 1 & 20 & 400 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 4 \\ 5 \\ 7 \\ 11 \\ 12 \\ 14 \\ 17 \\ 18 \\ 16 \\ 15 \\ 12 \end{bmatrix}$$

Orders of Different Matrices

Sl. No.	Matrix	Order
1	$\mathbf{X}$	
2	$\mathbf{X}'$	
2	$\hat{\beta}$ $[\hat{\beta}_0 \ \hat{\beta}_1 \ \hat{\beta}_2]'$	
3	$\mathbf{Y}$	
4	$\mathbf{X}'\mathbf{X}$	
5	$\mathbf{X}'\mathbf{Y}$	
6	$(\mathbf{X}'\mathbf{X})^{-1}$	

Please fill up the values in the above table

Orders of Different Matrices

Sl. No.	Matrix	Order
1	$\mathbf{X}$	11x3
2	$\mathbf{X}'$	3x11
2	$\hat{\beta}$ $[\hat{\beta}_0 \ \hat{\beta}_1 \ \hat{\beta}_2]'$	3x1
3	$\mathbf{Y}$	11x1
4	$\mathbf{X}'\mathbf{X}$	3x3
5	$\mathbf{X}'\mathbf{Y}$	$(3 \times 11) \times (11 \times 1) = 3 \times 1$
6	$(\mathbf{X}'\mathbf{X})^{-1}$	3x3

Parameter Estimation for II Order Model

$$\hat{\beta} = (X'X)^{-1}(X'Y)$$

$X'X =$

11	156	2324
156	2324	36216
2324	36216	587012

$(X'X)^{-1} =$

44.4288	-6.2551	0.2100
-6.2551	0.8918	-0.0303
0.2100	-0.0303	0.0010

$X'Y =$

131
1976
30820

$\hat{\beta} =$

-67.1947
10.2919
-0.3164

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X + \hat{\beta}_2 X^2$$

$$\hat{Y} = -67.195 + 10.292X - 0.3164X^2$$

Metrics of Regression

Sl. No.	Metrics	Metrics Formula	Value
1	Total Sum of Squares	$Y'Y$	1789
2		$\sum_{i=1}^{11} Y_i^2$	
3	Regression Sum of Squares	$\hat{\beta}'X'Y$	1781.768
4	Residual Sum of Squares	$(Y - X\hat{\beta})'(Y - X\hat{\beta})$	7.2316
5		$\sum_{i=1}^{11} (Y_i - \hat{Y}_i)^2$	
6	Total Sum of Residuals	$\sum_{i=1}^{11} (Y_i - \hat{Y}_i)$	0
7	Average Y	$\bar{Y} = \frac{\sum_{i=1}^{11} Y_i}{11}$	11.909
8	Average $\hat{Y}$	$\bar{\hat{Y}} = \frac{\sum_{i=1}^{11} \hat{Y}_i}{11}$	11.909

Matlab Code for Conveyor Belt Example

```
clear all; clc;
```

```
Data = [  
1 10 4  
2 10 5  
3 11 7  
4 12 11  
5 13 12  
6 14 14  
7 15 17  
8 16 18  
9 17 16  
10 18 15  
11 20 12  
];  
  
SINo = Data(:,1);  
Speed = Data(:,2);  
Luggages = Data(:,3);  
X = [ ones(numel(Speed),1) Speed Speed.^2];  
  
Y=Luggages;  
beta_hat = inv(X'*X)*(X'*Y);  
  
beta_hat =  
-67.1947  
10.2919  
-0.3164
```

```
Y_Pred = X*beta_hat;  
Compare = [SINo Y Y_Pred];
```

```
% Compare =  
%  
%      1.0000      4.0000      4.0807  
%      2.0000      5.0000      4.0807  
%      3.0000      7.0000      7.7275  
%      4.0000     11.0000     10.7414  
%      5.0000     12.0000     13.1224  
%      6.0000     14.0000     14.8706  
%      7.0000     17.0000     15.9859  
%      8.0000     18.0000     16.4684  
%      9.0000     16.0000     16.3180  
%     10.0000     15.0000     15.5347  
%     11.0000     12.0000     12.0696
```

% Fit seems real nice

Sum of Residuals and Total Sum of Squares

% Find average of Y and average of Y_Pred $\bar{Y}_{pred}$: 11.9091 (Both are same)

Average_Ys = [mean(Y) mean(Y_Pred)];

% What is sum of residuals? Pretty much zero

Residual_Sum=sum(Y-Y_Pred); % Pretty much zero

% What is total sum of squares : TSSQ

% Two ways - 1. Simply $Y'Y$ or 2. sum over Y^2_i

TtSSQ1 = $Y'Y$;

TtSSQ2 = sum($Y.^2$); % element by element squaring of terms

TtSSQ_Compare = [TtSSQ1 TtSSQ2];

% TSSQ_Compare = 1789 1789

Both methods gave same answer

Regression & Residuals Sum of Squares

% Let us find Regression SSQ RsSSQ1

RgSSQ1 = beta_hat'*X*Y; % 1781.8

% Let us find Residuals Sum of Squares by 2 methods 1.RsSSQ1 or 2. RsSSQ2

RssSQ1 = (Y-X*beta_hat)'*(Y-X*beta_hat);

RssSQ2 = sum((Y-Y_Pred).^2);

% Compare the two Residual SSQ

Compare_RsSSQ = [RsSSQ1 RsSSQ2];

% Compare_RsSSQ = [7.2316 7.2316] % Both approaches give the same Residual Sum of Squares

Sum of Squares Conservation

% Check if Total Sum of Squares =

% Regression Sum of Squares + Residual Sum of Squares

Total_SSQ = RgSSQ1+RsSSQ1; % Regression SSQ + Residual SSQ

TotalSSQ_Compare = [TtSSQ1 Total_SSQ]; % Both are 1789

% Regression Sum of Squares = 1781.7684;

% Residual Sum of Squares = 7.2316;

% Total of the above two = 1789.0

Optimization

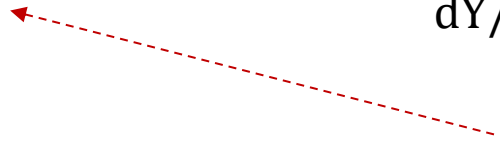
% Find optimal conveyor belt speed

% (differentiate Y w.r.t. X and set it to zero)

$$\hat{Y} = \hat{\beta}_1 + \hat{\beta}_2 X + \hat{\beta}_3 X^2$$

$$X_{\text{opt}} = -\left(\frac{1}{2}\right) \frac{\text{beta_hat}(2)}{\text{beta_hat}(3)}$$

$$d\hat{Y}/dX = 0 + \hat{\beta}_2 + 2\hat{\beta}_3 X$$

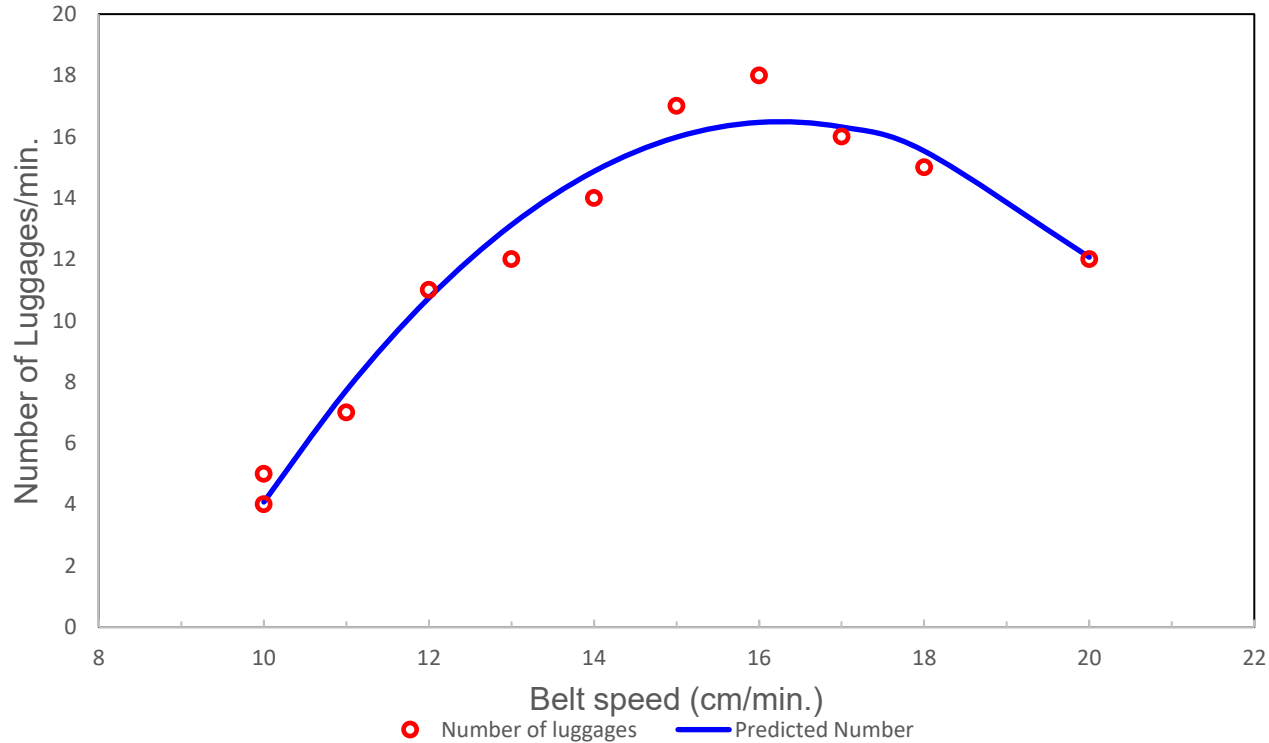
$$\hat{\beta}_2 + 2\hat{\beta}_3 X_{\text{opt}} = 0$$


$$y_{\text{opt}} = [1 \quad x_{\text{opt}} \quad x_{\text{opt}}^2] * \text{beta_hat};$$

$$x_{\text{opt}} = 16.26 \text{ (cm/s)}$$

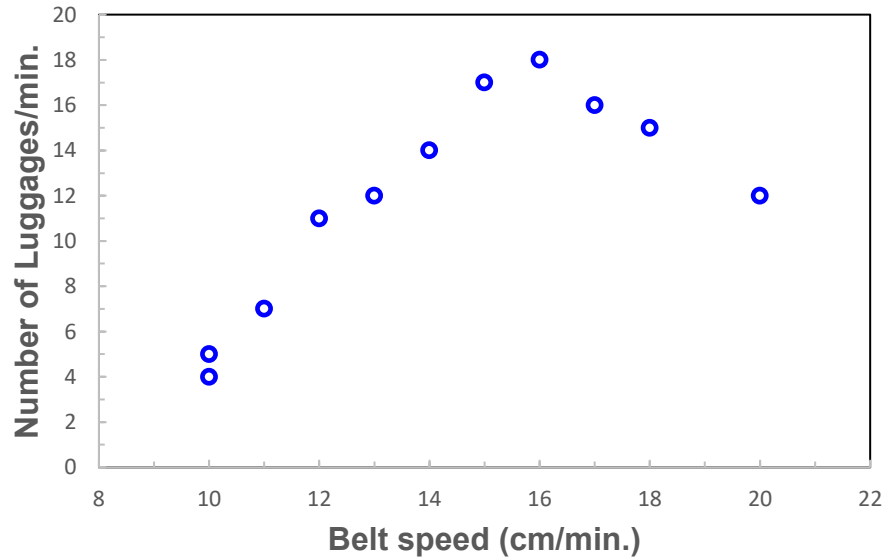
$$y_{\text{opt}} = 16.4902 \sim 16 \text{ luggages/minute}$$

Conveyor Belt Example: II Order Model Fit



Model Choice

First order model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X_1$ (obviously inadequate – may be compared with II order model)

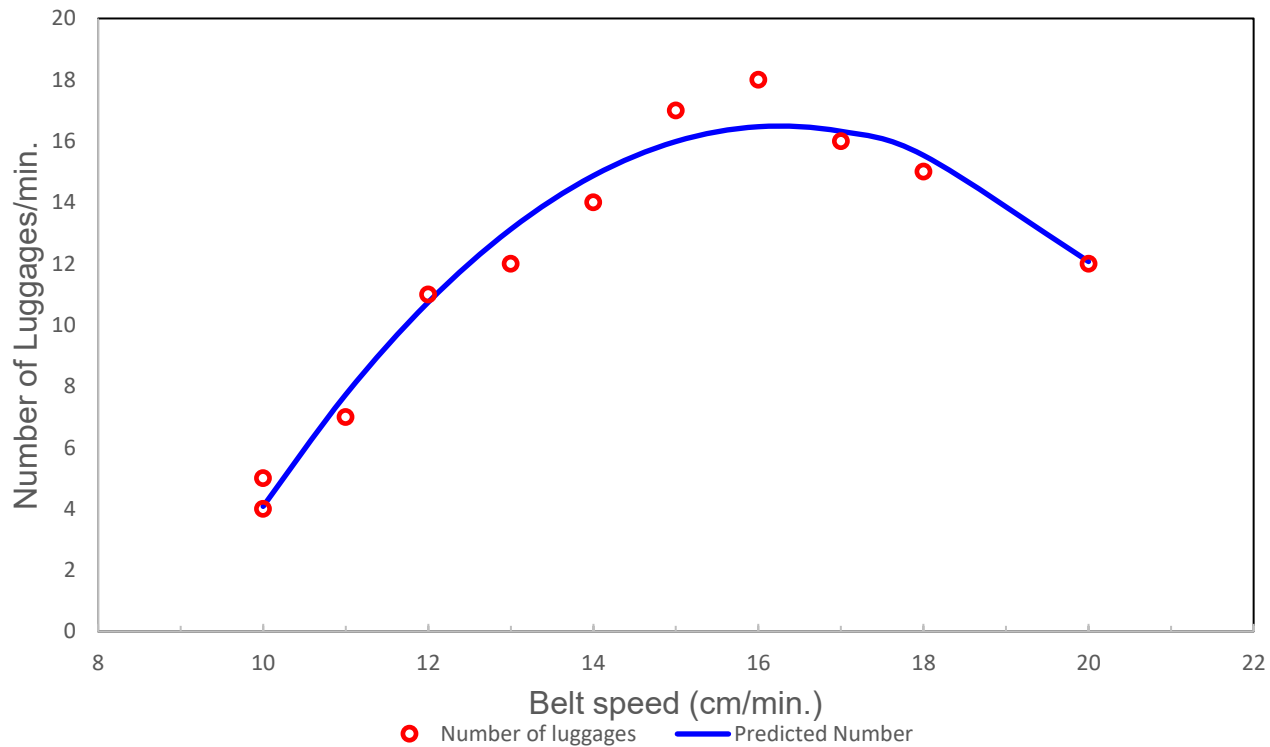


Comparing I and II Order Regression Models

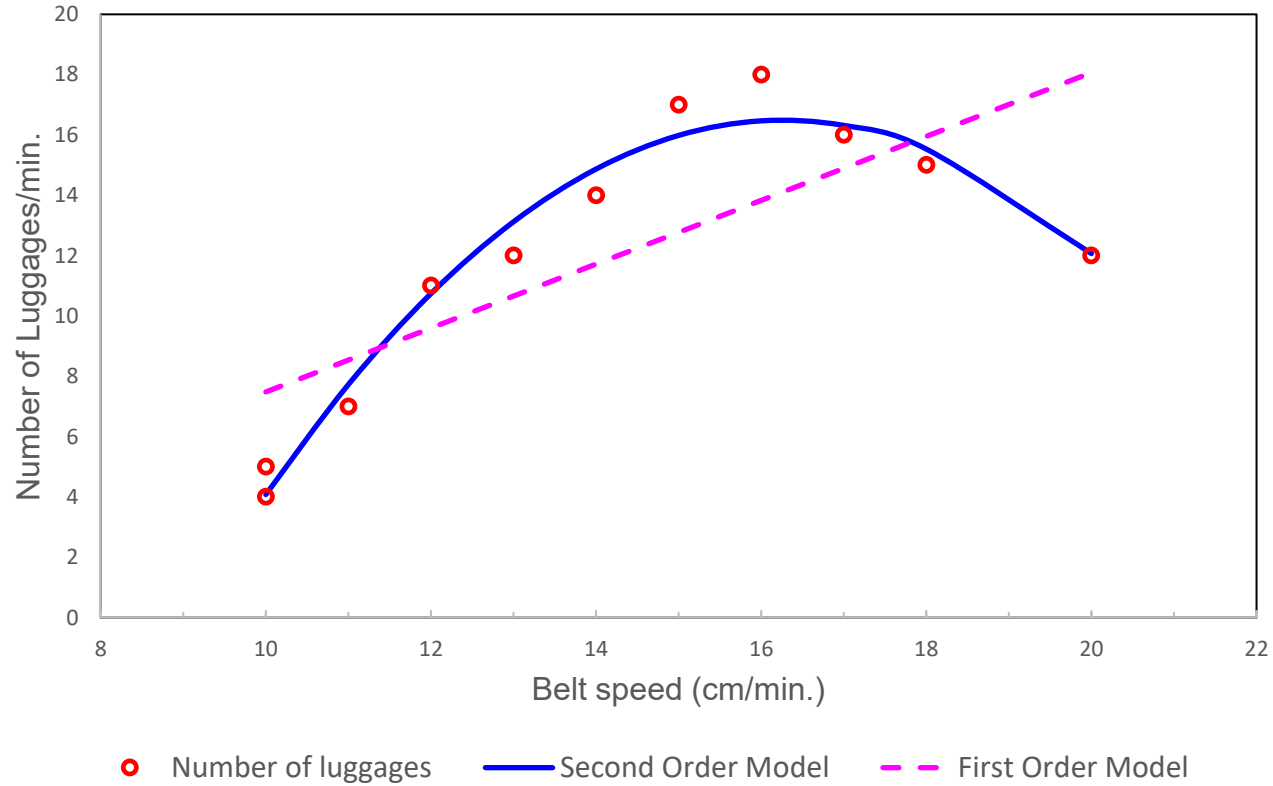
Sl. No.	Metrics	I order Model	II Order Model
1	Number of Parameters	2	3
3	Parameter Values	-3.1042 1.0586 $\hat{\beta}_0$ and $\hat{\beta}_1$	-67.1947 10.2919 -0.3164 $\hat{\beta}_0, \hat{\beta}_1$ and $\hat{\beta}_2$
6	Regression Sum of Squares	1685.202	1781.768
7	Residual Sum of Squares	103.798	7.232
8	Total Sum of Square	1789	1789

Conclusion: The II order model enjoys more Regression and less Residual Sum of Squares making it superior to the I order model

Conveyor Belt Example: II Order Model Fits



First & II Order Model Fits



Linear Regression in Matlab: 1 order Model

% Demonstrating polyfit in Matlab % TA may find equivalent code in Python

```
[pLinear,SLinear] = polyfit(Speed,Luggages,1)
```

```
eqLinear = "Linear: " + pLinear(1) + "x + " + pLinear(2)
```

```
xQuery = [0:1:20]';
```

```
yLinear = polyval(pLinear,xQuery);
```

```
% "Linear: -3.1042 + 1.0586x " 1 order model equation
```


Linear Regression in Matlab: II order Model

% Demonstrating polyfit in Matlab % TA may find equivalent code in Python

```
[pQuad,SQuad] = polyfit(Speed,Luggages,2)
```

```
eqQuad = "Quadratic: " + pQuad(1) + "x^2 + " + pQuad(2) + "x + " + pQuad(3)
```

```
xQuery = [0:1:20]';
```

```
yLinear = polyval(pLinear,xQuery);
```

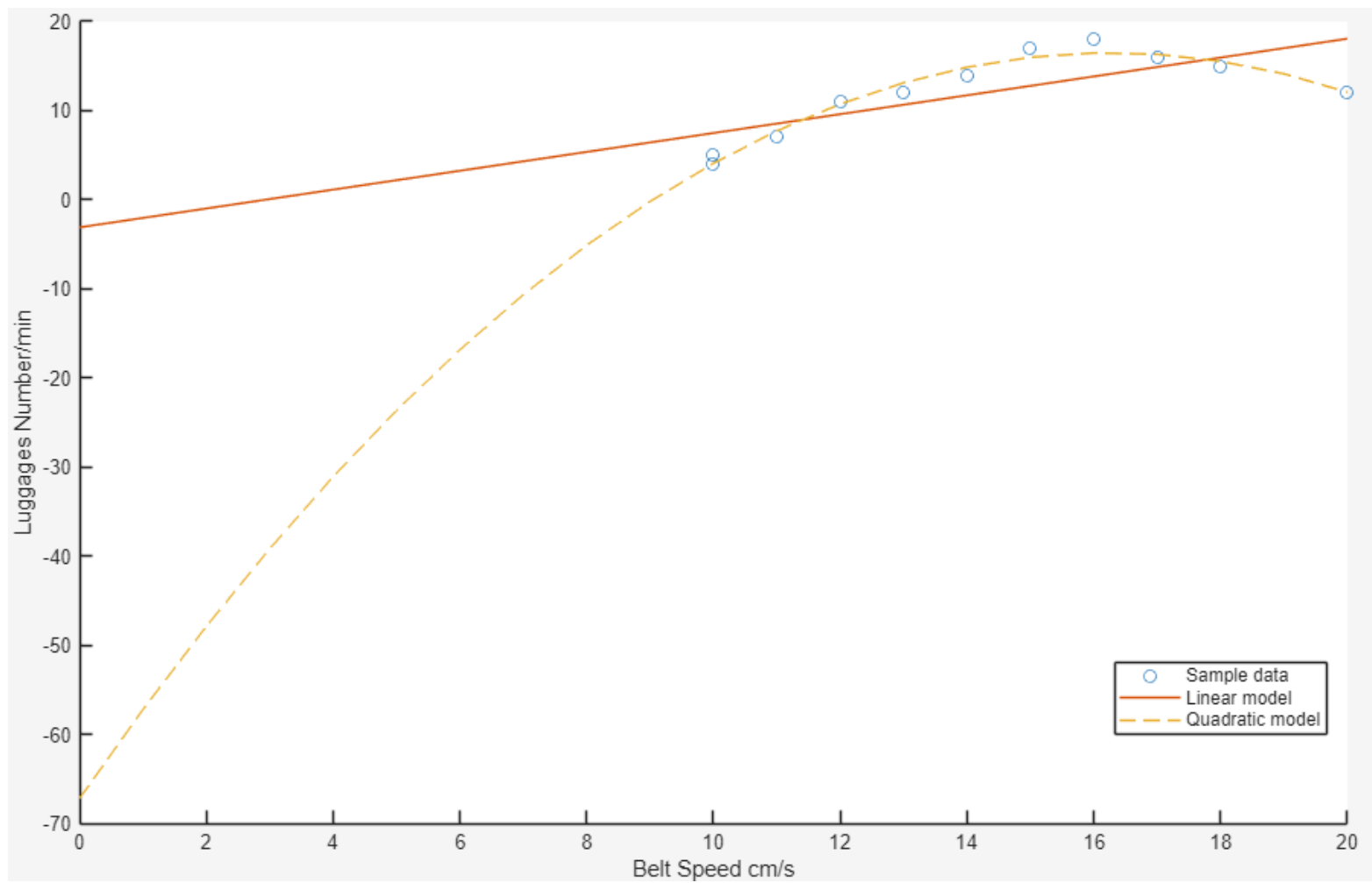
```
yQuad = polyval(pQuad,xQuery);
```

```
%
```

```
% "Quadratic: -67.1947 + 10.2919x + -0.31643x^2 : II order model
```

Linear Regression in Matlab: Plots

```
scatter(Speed,Luggages)
hold on
plot(xQuery,yLinear,"-")
plot(xQuery,yQuad,"--")
hold off
xlabel("Belt Speed cm/s")
ylabel("Luggages Number/min")
legend(["Sample data" "Linear model" "Quadratic model"])
text(0.3,30,[eqLinear eqQuad])
```



Multivariable Linear Regression in Matlab

Let us scale up the complexity from a simple linear regression model (toy model) involving

- a) Limited data (11 points)
- b) Single predictor (belt speed)
- c) Excellent Fit

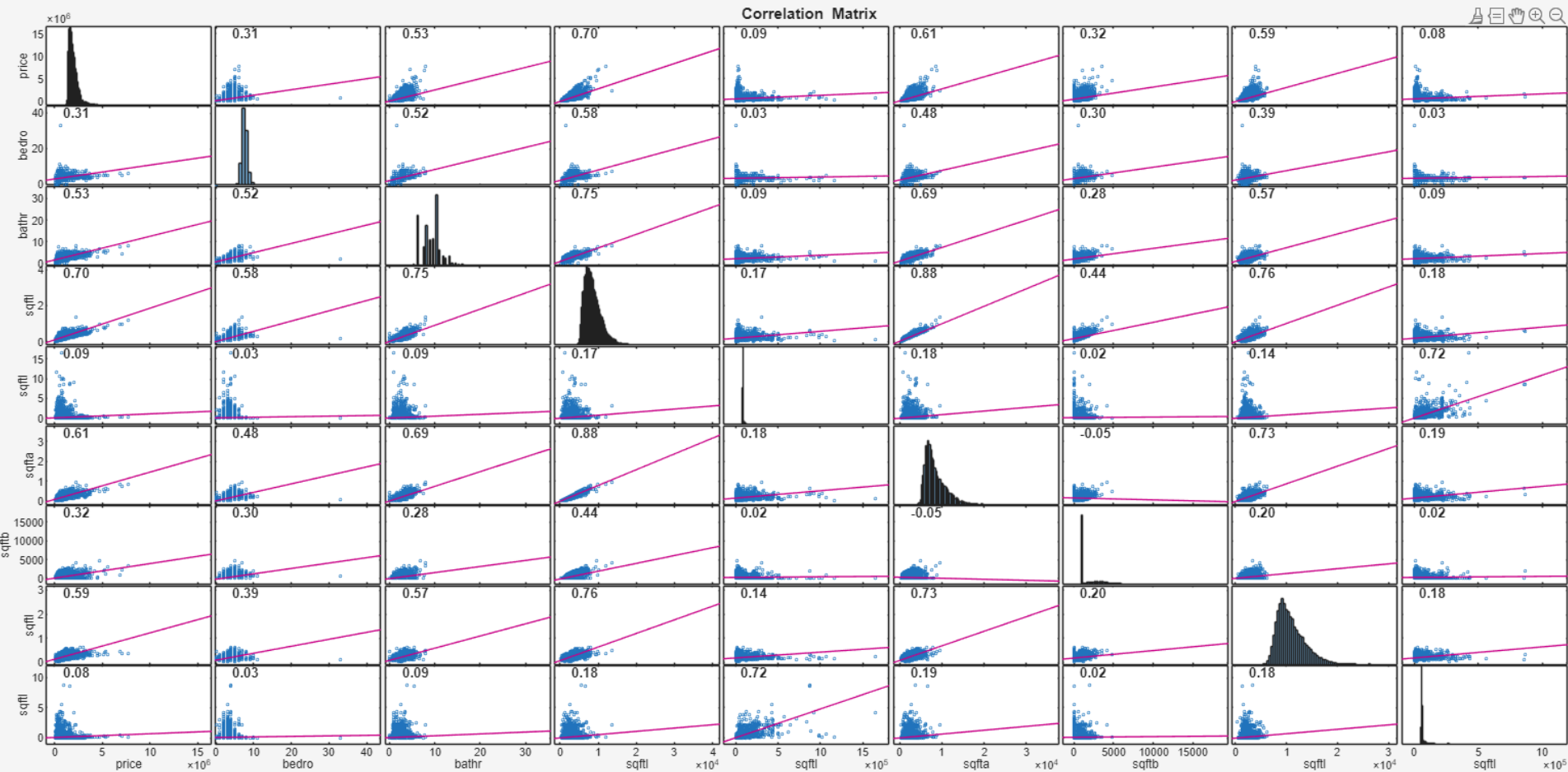
To

- c) Real life example
- d) Big data (21813 points)
- e) Multiple predictors (11) some of which may be strongly correlated
- f) May not get an excellent fit but only trends

Matlab Code For Housing Data Correlations

```
Tbl = array2table(Data, 'VariableNames', {'price', 'bedrooms',  
'bathrooms', 'sqftliving', 'sqftlot', 'sqftabove', 'sqftbasement',  
'sqftliving15', 'sqftlot15'});  
corrplot(Tbl);
```

Correlation Statistics with Matlab



Correlation Statistics from JMP

	price	bedrooms	bathrooms	sqft_living	sqft_lot	sqft_above	sqft_basement	sqft_living15	sqft_lot15
price	1.0000	0.3084	0.5251	0.7020	0.0897	0.6056	0.3238	0.5854	0.0824
bedrooms	0.3084	1.0000	0.5159	0.5767	0.0317	0.4776	0.3031	0.3917	0.0293
bathrooms	0.5251	0.5159	1.0000	0.7547	0.0877	0.6853	0.2838	0.5686	0.0872
sqft_living	0.7020	0.5767	0.7547	1.0000	0.1728	0.8766	0.4350	0.7564	0.1833
sqft_lot	0.0897	0.0317	0.0877	0.1728	1.0000	0.1835	0.0153	0.1446	0.7186
sqft_above	0.6056	0.4776	0.6853	0.8766	0.1835	1.0000	-0.0519	0.7319	0.1940
sqft_basement	0.3238	0.3031	0.2838	0.4350	0.0153	-0.0519	1.0000	0.2004	0.0173
sqft_living15	0.5854	0.3917	0.5686	0.7564	0.1446	0.7319	0.2004	1.0000	0.1832
sqft_lot15	0.0824	0.0293	0.0872	0.1833	0.7186	0.1940	0.0173	0.1832	1.0000

Let

Y=Price

X1 = bedrooms

X5 = sqft_above

X2 = bathrooms

X6 = sqft_basement

X3 = sqft_living

X7 = sqft_living15

X4 = sqft_lot

X8 = sqft_lot15

Normalizing Matlab Code For Housing Data

As the predictors data (see a typical line below) were in different orders of magnitudes we first normalize the data using Z-score

1 940 5000 940 0 1560 4500

$$Z = \frac{(X - \text{mean}(X))}{\sigma(X)}$$

Z = Normalized score $\text{mean}(X)$ = average of Feature X

$\sigma(X)$ = standard deviation of Feature X

Normalizing Matlab Code For Housing Data

$$Z = \frac{(X - \text{mean}(X))}{\sigma(X)}$$

Z = Normalized score $\text{mean}(X)$ = average of Feature X

$\sigma(X)$ = standard deviation of Feature X

As a result of standardization all features are made to have a mean of 0 and standard deviation of unity

	X1	X2	X3	X4	X6	X7	X8
Mean	-6.33392E-16	4.5E-16	5.13E-16	-5.8E-17	-6.9E-15	1.66E-16	-3.8E-16
STD	1	1	1	1	1	1	1

Normalizing Housing Data Features

% The data in the code is first relabelled to X1, X2,...X8 for the 8

% predictors and Y for price

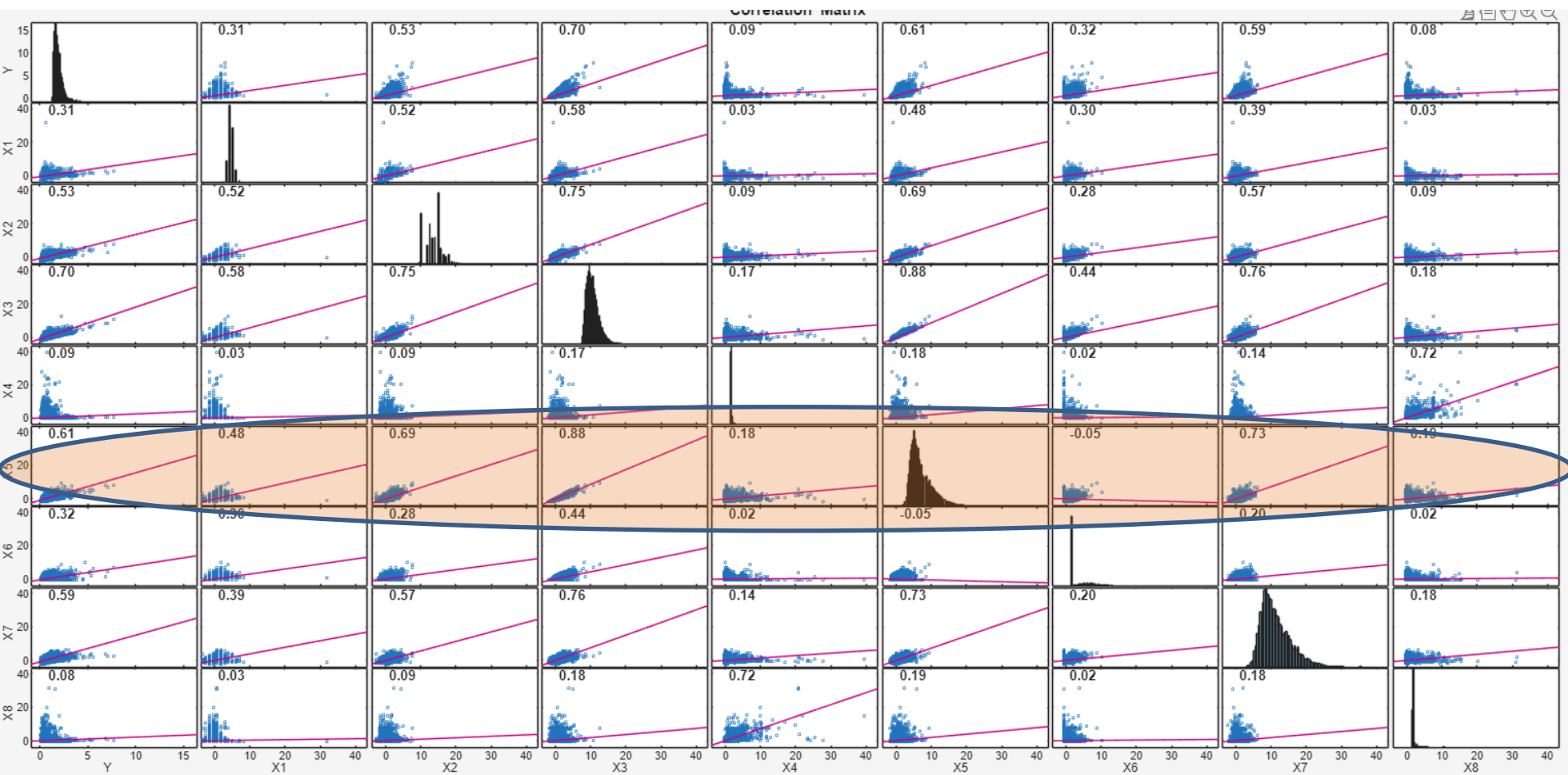
```
Features = [X1, X2, X3, X4, X5, X6, X7, X8];
```

```
Target = Y/1e6; % Express house price in millions of dollars
```

```
Data_Norm = [ Target normalize(Features)];
```

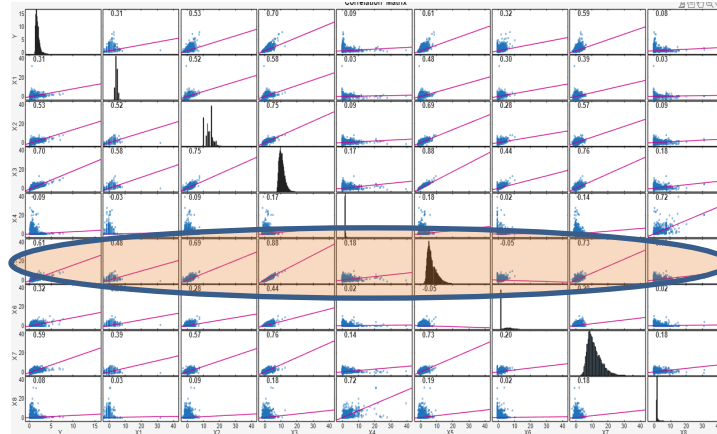
```
Tb2 = array2table([Data_Norm], 'VariableNames', {'Y', 'X1', 'X2', 'X3', 'X4',  
, 'X5', 'X6', 'X7', 'X8'});
```

Correlation For Normalized Data with Matlab: No Change



Data Analysis

- ❖ As shown in the figure in previous slide, we find X5 has strong correlation with most features and probably removing it may not affect the information content in the data
- ❖ Retaining X5 predictor may lead to difficulties in parameter estimations

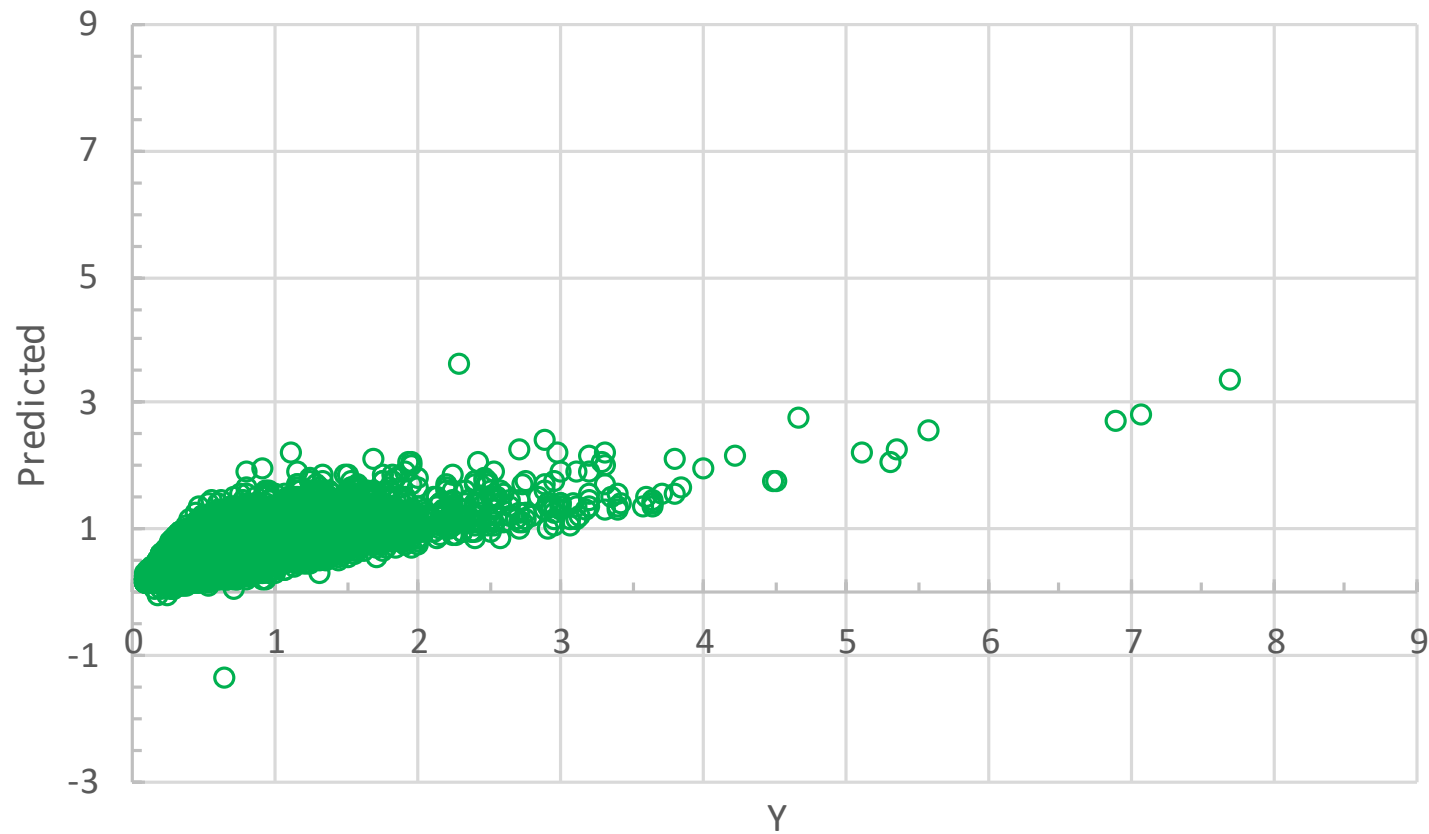


Running Linear Regression

% Linear regression was run as before and % we get the parameters as

Coefficient of	Normalized Feature	Parameter	Value
Intercept		$\hat{\beta}_0$	0.5402
X1	# of bedrooms	$\hat{\beta}_1$	-0.0545
X2	# of bathrooms	$\hat{\beta}_2$	0.0057
X3	sqft_living	$\hat{\beta}_3$	0.2442
X4	sqft_lot	$\hat{\beta}_4$	0.0018
X6	sqft_basement	$\hat{\beta}_6$	0.0182
X7	sqft_living15	$\hat{\beta}_7$	0.0489
X8	sqft_lot15	$\hat{\beta}_8$	-0.0240

Inspecting Model Fit



Analysis

- ❖ Normalizing the data and scaling the target brought the numbers to more or less consistent range for all predictors and outcome
- ❖ With the estimated regression coefficients we find square feet of the house having maximum impact on the costs
- ❖ Had we included X_5 in the model, the matrix inversion would have been difficult as the resulting matrix was nearly singular indicating strong collinearity between certain features
- ❖ The fit may definitely be improved
- ❖ There seemed to be only a couple of outliers in the data which could not have compromised the fit seriously